The Socioeconomic Burden on Your Heart

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***Abstract*—This paper explores how socioeconomic factors— such as educational attainment and retirement benefits—influence heart attack rates across counties in Wisconsin. We utilized regression techniques determining statistically significant negative correlations, indicating that lower higher education attainment rates (p = 6.23E-7) and lower retirement benefits (p = 0.002), are linked to higher heart attack rates. We hope our findings will encourage the development of targeted public policies and initiatives. Specifically, promoting greater access to higher education and enhancing retirement benefits, such as increasing standard wages or pension contributions, may effectively reduce socioeconomic stressors, ultimately improving cardiovascular health outcomes and community longevity.**

I. Introduction

How do you stop an ailment with nearly unlimited causes? One so pervasive that it afflicts more than 800.000 individuals in the United States within a year (NIH, 2022). Heart attacks all start with a lack of oxygen present in heart muscle due to arterial blockage. It can leave the individual with chest pain, shortness of breath, uncontrollable sweating, exhaustion, rapid heartbeat, and sudden dizziness, or no symptoms at all (Mayo Clinic, 2023). However, humanity continues to learn how to subdue this sudden killer in the hopes of protecting our future.

A. *Background and Related Literature*

The Mayo Clinic (2023) states that the leading cause of heart attacks is due to coronary heart disease. This is when one of the heart's arteries are blocked due to cholesterol deposits called plaques (American Heart Association, 2021). The plaques prohibit the entry of oxygen rich blood into the heart, increasing the probability of a heart attack. The stress hormone Cortisol further inflames the heart increasing the risk of rupturing a plaque (American Heart Association, 2021). The Multi-Ethnic Study of Atherosclerosis, specifically the sub-study “MESA Stress 1” highlighted increased cortisol levels can lead to a 90% increase in heart attack events (American Heart Association, 2021).

Habibov et al. (2019) demonstrated that poverty leads to deteriorating health quality, which is directly linked to an increased risk of heart attacks. Poverty also limits a community’s access to education, both through reduced funding and because personal hardships often take precedence (Buck and Deutsch, 2014). Education opens doors to more fulfilling employment, characterized by less repetitive tasks and continuous learning opportunities, along with increased economic resources, all factors linked to reduced psychological distress (Ross & Van Willigen, 1997). Similarly, Yeung and Zhou (2017) demonstrate that effective retirement planning significantly reduces stress during the initial year of retirement, reinforcing the importance of stable socioeconomic support systems in maintaining cardiovascular health.

B. *Scope of the Study*

This paper examines the influence of socioeconomic factors on heart attack rates among Wisconsin residents aged 35 and older. We hypothesized that more favorable socioeconomic conditions would correspond with lower crude heart attack rates. We analyzed two socioeconomic independent variables, educational attainment percentages and retirement benefit dollars against the dependent variable county-level heart attack rates.

II. Methods

We analyzed data from 2017–2019 to minimize external factors that could distort our results. Specifically, this period was selected to avoid the economic instability associated with major events like the 2008 housing market crash and the COVID-19 pandemic, ensuring a clearer representation of typical socioeconomic conditions.

A. *Compiling the Data*

Using CDC data, we obtained the crude hospitalization rate for heart attacks in Wisconsin counties as our dependent variable. We adjusted age and population size by focusing on individuals over 35 to minimize confounding factors related to extenuating circumstances common in younger populations such as congenital heart defects (Gilboa e al., 2010). Additionally, collecting admissions data from state and local hospitals provided a more immediate view of case severity compared to discharge data. The crude heart attack rate data includes information ranging from the 2000s to 2021,

FRED, run by the Research Department at the Federal Reserve Bank of St. Louis collects and condenses data sets onto one platform from private and public sectors. FRED offers higher education attainment rates per Wisconsin county from 2010 onwards, allowing a wide range of data for our analysis. FRED's automatic CSV generator outputs data requiring complex REGEX parsing to match tags from a README file. To avoid this complexity, we manually collected data.

We aim to determine whether government preparation and benefits for retirement can reduce an individual’s stress and therefore reduce the probability of a heart attack. The Social Security Administration provides information on the quantity of money OASDI (Old-Age, Survivors, and Disability Insurance) was given in thousands of dollars. The data was manually gathered from the administration's yearly reports.

B. Procedural Framework

We chose to use python, (specifically a Jupyter Notebook) to compile the data in this paper. The standard imports for all the following code are:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats
import statsmodels.api as sm
```

Each data set was compiled into separate CSV files. The heart attack rates were then compared against each CSV. Data cleaning was necessary, including formatting adjustments, replacing county category names, selecting relevant years to prevent skewed results or mismatched data. The data was then averaged over the selected years for each county to provide a more stable representation and smooth out an outlier. The scipy library was utilized to calculate correlation and p-values, minimizing human error in statistical analysis. Additionally, a statsmodels linear regression model using least squares was trained to further analyze relationships within the data. Figure 1 at the end of the document illustrates the respective code used in the processes.

III. Analysis

The Ordinary Least Squares (OLS) model from the **statsmodels** library provides extensive information regarding the relationship between independent variables and the dependent variable, heart attack rates. However, for brevity, this paper will primarily focus on three key statistical measures: the Pearson correlation coefficient (r), R-squared value, and regression coefficients. The null hypothesis proposed that no statistically significant relationship existed between any of the examined independent work-life factors and heart attack rates."

A. Higher Education Lower Risk

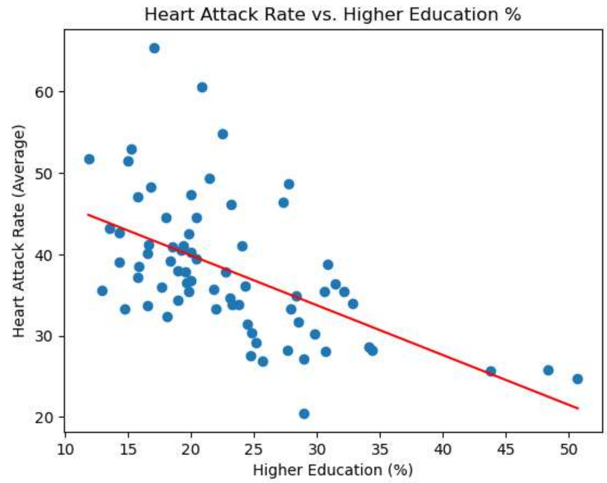

Figure 3. Scatter plot depicting the relationship between the average percentage of residents with a bachelor's degree and the average heart attack rate by county (2017–2019). Each point represents a county in Wisconsin. The red line indicates a linear regression fit.

OLS Regression Coefficients						
	Coefficient	Standard Error	t	P > |t|	[0.025	0.975]
Constant	52.0745	2.711	19.207	0.000	46.666	57.483
Bachelors Rate	-0.6118	0.111	-5.491	0.000	-0.834	-0.390

Table 2a. OLS regression coefficient output generated using Python's **statsmodels** library. This table displays the y-intercept (Constant) and slope for the independent variable (Bachelor Rate) after fitting the model to the data. "Bachelors Rate" refers to the name of the column in the pandas DataFrame representing higher education attainment rates. The name is a remnant of the code's implementation and does not affect interpretation of the results.

OLS Model Summary	
Statistic	**Value**
Independent Variable	Higher Education Attainment Rate
Dependent Variable	Heart Attack Rates
Pearson Correlation (r)	-0.551443
P-value (Correlation)	6.2297E-07
R-squared	0.304

Table 2b. Summary output comparing Higher Education Attainment vs. Heart Attack Rates of an Ordinary using a Least Squares (OLS) regression model using the **statsmodels** library in Python. After adding a constant term to account for the interception.

Figure 3 and table 2 show a significant negative correlation between bachelor's degree attainment and heart attack rates ($p \approx 6.23\times10^{-7}$). The OLS regression explains about 30% of this variation, indicating heart attack rates decrease by roughly 0.61 units per 1% increase in higher education levels (y-intercept ≈ 52.07). Thus, higher education rates are significantly associated ($p < 0.05$) with lower heart attack rates in this sample.

B. Retirement Benefits and Cardiovascular Health

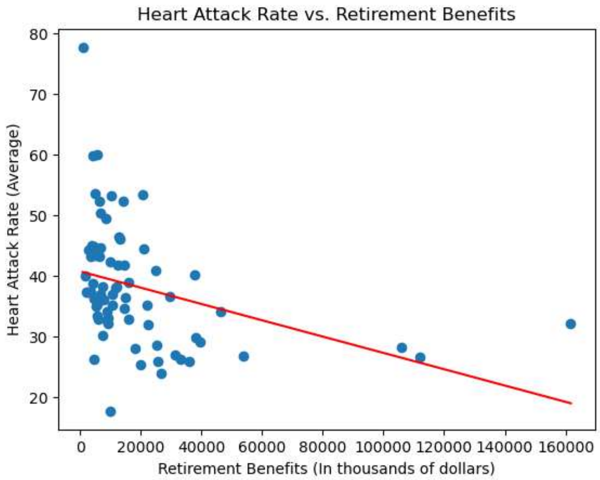

Figure 4. Scatter plot showing the relationship between average retirement benefits and average heart attack rates by county (2017–2019). Each point represents Wisconsin county. The red line represents a fitted linear regression.

OLS Regression Coefficients						
	Coefficient	**Standard Error**	**t**	**P > |t|**	**[0.025**	**0.975]**
Constant	40.7864	1.364	19.910	0.000	38.067	43.506
Retirement Benefits	-0.0001	4.19E-05	-3.136	0.003	-0.000	-4.9E-05

Table 3b. OLS regression coefficient output generated using Python's **statsmodels** library. This table displays the estimated intercept (constant) and slope for the independent variable (Retirement Benefits) after fitting the model with a constant term added using *sm.add_constant()*.

OLS Model Summary	
Statistic	**Value**
Independent Variable	Retirement Benefits
Dependent Variable	Heart Attack Rates
Pearson Correlation (r)	-0.350964
P-value (Correlation)	0.0025053
R-squared	0.123

Table 3a. Summary output comparing Retirment Benefits vs. Heart Attack Rates of an Ordinary using a Least Squares (OLS) regression model using the **statsmodels** library in Python. After adding a constant term to account for the y-intercept.

The results indicate a weaker but statistically significant negative relationship between retirement benefits and heart attack rates with $r \approx –0.351$ and $p \approx 0.0025$. The OLS model explains about 12% of the variation. The regression slope (–0.0001, $p = 0.003$) suggests that areas with higher retirement benefits tend to have slightly lower heart attack rates on average. By our biology lab's standard of ($p < 0.05$), we can conclude there exists a correlation between higher retirement benefits and lower heart attack rates.

IV. Discussion

Based on the statistical analysis performed, we reject the null hypothesis. Our findings show that both higher education attainment and more robust retirement benefits are each significantly and negatively correlated with heart attack rates, indicating that lower socioeconomic status increases the likelihood of experiencing a heart attack.

However, while our findings suggest education and retirement aid influence heart attack rates, there is still unaccounted variance. These could be explained through alternate life factors such as healthcare, diet, or exercise. One such example is in long-term cohort study of San Francisco longshoremen, manual laborers who load and unload ship cargo, found that those engaging in higher levels of occupational physical activity had approximately half the risk of fatal heart attack compared to those in the lowest activity group, even after adjusting for numerous health and lifestyle factors (Brand et al., 1971). Brand et al. (1971), suggests factors such as physical labor may mitigate the adverse effects; however, we propose that due to the location on the west coast these longshoremen would be a part of the International Longshore and Warehouse Union-Pacific Maritime Association pension plan (*ILWU*, 2017). As we demonstrated, government retirement aid lowers heart attack rates, this can be extended to union retirement plans. Therefore, further research isolating and uniformly applying the extenuating circumstances can be conducted to potentially capture more variance.

A. Areas for Further Research

One potential limitation from the heart attack rate data is that there is no differentiation between sexes. As Howard (2018) discussed in her article, women's symptoms of heart attacks are less recognizable than men. Therefore, the data could be skewed to those with easier access to healthcare as those with stricter insurance may be hesitant to use it without proper assurance of a heart attack.

Furthermore, when considering retirement benefits, woman may disproportionally be stressed as stated by Bill McInturff, "Nearly 6 in 10 (59%) women and 81% of low-income women told us they do not make enough money right now to save for retirement" (NCOA, 2024). Therefore, a gender specific study may reveal a more complex relationship between work life balance and heart attack rates.

For Heart Attack Rates versus retirement benefits, the data we looked at heavily skewed to the left: most points cluster around lower benefit levels, with only a handful extending far to the right. This skewness can make the regression line sensitive to those few extreme observations. Further analysis of retirement benefits could use more robust regression techniques such as Huber regression or RANSAC to reduce the impact of outliers, and transformations or non-linear models might also better capture the distribution's skew.

*B. Moving Fo*rward

Our research demonstrated that current environmental factors in the United States significantly impact heart health. Recognizing this connection allows us to propose targeted solutions addressing socioeconomic imbalances, ultimately raising wellbeing both socially and biologically. One plausible solution is implementing a standard nationwide wage enabling individuals a holistic, fulfilling life rather than covering survival expenses reducing the incidence of poverty. Strengthening retirement benefits would alleviate financial insecurity among older adults, reducing chronic stress and its cardiovascular toll. Likewise, promoting the benefits of secondary education to adolescents from an early age can boost higher-education attainment and further mitigate poverty. These combined efforts could significantly contribute to reducing heart disease, one of the leading causes of death in the United States.

```python
### Grabbing Data from CSVs ###
heart_df = pd.read_csv('HA_rates_2000_2021.csv')
edu_df   = pd.read_csv('higher_education_2017_2019.csv')

### Cleaning Data ###
heart_df['Value'] = pd.to_numeric(heart_df['Value'], errors='coerce')
heart_df['Year'] = pd.to_numeric(heart_df['Year'], errors='coerce')

heart_df.dropna(subset=['Value'], inplace=True)
edu_df.dropna(subset=['Percent'], inplace=True)

heart_df = heart_df[(heart_df['Year'] >= 2017) & (heart_df['Year'] <= 2019)]
edu_df = edu_df[(edu_df['Year'] >= 2017) & (edu_df['Year'] <= 2019)]

heart_df.rename(columns={'Value': 'HeartAttackRate'}, inplace=True)
edu_df.rename(columns={'Percent': 'BachelorsRate'}, inplace=True)

edu_df['County'] = edu_df['County'].str.replace(' County, WI', '')

### Averaging Data by County ###
heart_by_county = heart_df.groupby('County', as_index=False)['HeartAttackRate'].mean
    ()
edu_by_county = edu_df.groupby('County', as_index=False)['BachelorsRate'].mean()

### Merging on County ###
merged_df = pd.merge(heart_by_county, edu_by_county, on='County', how='inner')

### STATS ###
pearson_r, p_val = stats.pearsonr(merged_df['BachelorsRate'], merged_df['
    HeartAttackRate'])
print("Pearson Correlation (r):", pearson_r)
print("P-value:", p_val)

# LINEAR REGRESSION
#     Independent variable: Higher Education Rate
#     Dependent variable: Heart Attack Rate
X = merged_df['BachelorsRate']
y = merged_df['HeartAttackRate']
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())

### Plotting Data ###
plt.scatter(merged_df['BachelorsRate'], merged_df['HeartAttackRate'])
plt.xlabel("Higher Education (%)")
plt.ylabel("Heart Attack Rate (Average)")
plt.title("Heart Attack Rate vs. Higher Education %")

x_vals = np.linspace(merged_df['BachelorsRate'].min(), merged_df['BachelorsRate'].
    max(), 100)
x_vals_const = sm.add_constant(x_vals)
y_vals = model.predict(x_vals_const)
plt.plot(x_vals, y_vals, color='red')

plt.show()
```

Fig. 1a. Python script for performing linear regression analysis on average heart attack rates and higher education attainment rates by Wisconsin counties using data from 2017–2019. The code cleans and filters both datasets, aggregates by county, merges them, calculates the Pearson correlation, fits an OLS model with an intercept, and visualizes the results with a scatter plot and regression line.

```python
### Grabbing Data from CSVs ###
heart_df = pd.read_csv('HA_rates_2000_2021.csv')
benefits_df = pd.read_csv('retirement_benefits_2017_2019.csv')

### Cleaning Data ###
heart_df['Value'] = pd.to_numeric(heart_df['Value'], errors='coerce')
heart_df['Year'] = pd.to_numeric(heart_df['Year'], errors='coerce')

heart_df.dropna(subset=['Value'], inplace=True)
benefits_df.dropna(subset=['Retirement_Benefits'], inplace=True)

heart_df = heart_df[(heart_df['Year'] >= 2017) & (heart_df['Year'] <= 2019)]
benefits_df = benefits_df[(benefits_df['Year'] >= 2017) & (benefits_df['Year'] <=
    2019)]

heart_df.rename(columns={'Value': 'HeartAttackRate'}, inplace=True)

benefits_df = benefits_df[benefits_df['County'] != "Total, Wisconsin"]

### Averaging Data by County ###
heart_by_county = heart_df.groupby('County', as_index=False)['HeartAttackRate'].mean
    ()
benefits_by_county = benefits_df.groupby('County', as_index=False)['
    Retirement_Benefits'].mean()

### Merging on County ###
merged_df = pd.merge(heart_by_county, benefits_by_county, on='County', how='inner')

### STATS ###
pearson_r, p_val = stats.pearsonr(merged_df['Retirement_Benefits'], merged_df['
    HeartAttackRate'])
print("Pearson Correlation (r):", pearson_r)
print("P-value:", p_val)

# LINEAR REGRESSION
#    Independent variable: Retirement Benefits
#    Dependent variable: Heart Attack Rate
X = merged_df['Retirement_Benefits']
y = merged_df['HeartAttackRate']
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())

### Plotting Data ###
plt.scatter(merged_df['Retirement_Benefits'], merged_df['HeartAttackRate'])
plt.xlabel("Retirement Benefits (In thousands of dollars)")
plt.ylabel("Heart Attack Rate (Average)")
plt.title("Heart Attack Rate vs. Retirement Benefits")

x_vals = np.linspace(merged_df['Retirement_Benefits'].min(), merged_df['
    Retirement_Benefits'].max(), 100)
x_vals_const = sm.add_constant(x_vals)
y_vals = model.predict(x_vals_const)
plt.plot(x_vals, y_vals, color='red')

plt.show()
```

Fig. 1b. Python script performing an OLS regression analysis between average heart attack rates and retirement benefits by Wisconsin county. The code filters the dataset to the years 2017–2019, cleans and merges heart attack and retirement data, computes a Pearson correlation, fits a regression model with an intercept, and visualizes the results with a scatter plot and fitted regression line.

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Appendix

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